Roshan P. Patel

Dayton, OH 45440 | 937-602-6347 | rpatel968@gatech.edu | US Citizen | linkedin.com/in/roshanpatelgt

Available to intern **Winter/Spring 2027** (Jan-May), **Summer 2027** (May - August)

EDUCATION

Georgia Institute of Technology | Atlanta, GA **August 2026 – Present**
B.S. Computer Engineering Expected Graduation, May 2028

SKILLS

Programming: Java, Python, C++, MATLAB

Tools & Platforms: Git, GitHub, PyTorch, MATLAB, ROS2

Technical Concepts: Kalman Filtering and State Estimation, Software-in-the-Loop Testing, Backpropagation, Computer Vision, Data Structures and Algorithms, Debugging, Networking

Relevant Coursework: Object-Oriented Programming, Data Structures & Algorithms, Linear Algebra, Physics I & II

EXPERIENCE

Yellow Jacket Space Program | Atlanta, GA **September 2026 - Present**
Avionics Software Team, Guidance Navigation and Control System

- Own the magnetometer measurement model in the team's **MATLAB** software-in-the-loop simulation of a **22-state extended Kalman filter**, translating datasheet specifications into sensor behavior (noise, quantization, saturation, thermal drift) validated across a **105K-sample simulated flight**
- Uncovered a **33% underestimate** in magnetometer noise by comparing firmware configuration against datasheet specifications, revealing the filter's assumed measurement noise was **~4× too optimistic**
- Diagnosed a framework defect that silently ran sensor models noise-free (invalidating filter testing), and modeled Earth's magnetic field along the 3D trajectory via the World Magnetic Model at **~95× lower compute cost**

Georgia Tech Extended Reality | Atlanta, GA **August 2026 – Present**
Robot Dog Team

- Implementing locomotion control and position estimation on a Unitree Go2 quadruped robot, and deploying pre-trained reinforcement learning policies (gait, balance behaviors) onto physical hardware
- Working within a ROS2-based robotics stack to bridge trained AI models to connect to NVIDIA Jetson Orin Nano

Verimint Software Solutions | Dayton, OH **April 2026 – August 2026**
Co-Founder, COO

- Built a software company from the ground up to serve small businesses utilizing client negotiations, and sales
- Built full-stack web applications for small-business clients using an Agile workflow, shipping in short iterations driven by client feedback to deliver **50+ products** for **10+ local businesses**

PROJECTS

Autodifferentiation Engine & Neural Network Library | Independent Project **July 2026 – Present**

- Implemented a scalar autograd engine from scratch in Python using operator overloading and a dynamic computational graph, with backpropagation via topological sort and chain-rule gradient accumulation
- Verified gradients against PyTorch autograd, then extended the engine with new operations and refactored gradients
- Trained a multilayer perceptron end-to-end on the engine, implementing the loss function and gradient descent updates

Personal Planning Dashboard | Independent Project **March 2026 - June 2026**

- Built a productivity dashboard from scratch using HTML, and CSS having **8 pages** and **20+ collapsible sections**
- Created a custom system to track and update **146+ items** across multiple areas with real-time progress tracking and a pinned section pulling data from multiple sources

ACTIVITIES

- Vertically Integrated Project: Agentic AI Research - In Process of Publishing Paper
- Society of Asian Scientists & Engineers
- Georgia Tech Men's Basketball Assistant Director of Student Engagement